Prevalence and spatial distribution of the ectoparasites on the gills of Mystus vittatus *from River Penna flowing through YSR District, Andhra Pradesh, India*

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ONLINE FIRST

Journal of Parasitic Diseases

ISSN 0971-7196

J Parasit Dis
DOI 10.1007/s12639-020-01275-9

J Parasit Dis
https://doi.org/10.1007/s12639-020-01275-9

ORIGINAL ARTICLE

Prevalence and spatial distribution of the ectoparasites on the gills of *Mystus vittatus* from River Penna flowing through YSR District, Andhra Pradesh, India

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Received: 3 April 2020 / Accepted: 12 September 2020

Abstract A parasitic survey was performed during July 2017 to February 2018 to emphasize the prevalence and spatial distribution of ectoparasites in *Mystus vittatus* from three sampling sites of River Penna flowing through YSR District, Andhra Pradesh by routine parasitological procedures. A total of 238 ectoparasites were obtained from the gills of 70 examined fishes with a prevalence of 77.1%, mean intensity of 4.4 and mean abundance of 3.4 respectively. Three species of ectoparasites i.e., two monogeneans—*Cornudiscoides vittati* (n = 119); *Bifurcohaptor indicus* (n = 90) and one copepod, *Lamproglena hospetensis* (n = 29) were obtained from the gills of the host. All the 3 species of gill ectoparasites showed an aggregated distribution pattern (2.92, 2.52 and 2.83). Only, *B. indicus* showed a significant positive relationship between the condition factor and parasitic abundance ($r_s = 0.361$, $p = 0.0021$) followed by *C. vittati* ($r_s = 0.206$, $p = 0.086$) whereas the parasitic abundance of *L. hospetensis* ($r_s = -0.213$, $p = 0.076$) showed a weak negative correlation with the relative condition factor of the host. Parasitization was analyzed location wise and fishes collected from Site-I (Mylavaram) were highly infected with both the monogeneans while parasitization with copepods was nil. However, the parasitization of copepods was relatively higher in Site-II (Chennur). There was no influence of host size and host sex on the parasitization and also no specific preference for the sides of the fish host. The monogeneans segregated the niche to prevent resource competition i.e., *C. vittati* and *L. hospetensis* were prevalent in gill arches II and I respectively whereas *B. indicus* occupied gill arch III. Both the monogeneans showed very a high interspecific association ($r_j = 0.96$) with each other but showed a very negligible association with *L. hospetensis*. This type of study will help the aquaculturists to implement many advanced management strategies in aquaculture and improve the yield of *Mystus vittatus* in the Southern states of India.

Keywords *Cornudiscoides vittati* · *Bifurcohaptor indicus* · *Lamproglena hospetensis* · *Mystus vittatus* · Spatial distribution

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Introduction

Mystus vittatus (striped dwarf catfish) commonly known as 'tengra' in India is a species of catfish family: Bagridae mostly inhabiting the brackish water systems with marginal vegetation in lakes and swamps of Asian countries such as India, Pakistan, Sri Lanka, Nepal, Bangladesh and probably Myanmar and reaches a length of 21 cm in maximum (Froese and Pauly 2011; Vankara and Chikkam 2017). These fish serve as a chief source of animal protein and micronutrients in the diet of people (Ahmed et al. 2012; Vankara and Chikkam 2017). However, their population is known to be declining in the recent past, due to overfishing, pet trading, parasitic infection, and habitat destruction (Ng 2010). Fish gills serve as the most preferred habitats of many external parasites such as monogeneans, certain groups of parasitic copepods and isopods, which can be extremely numerous (Kearn 2004). These parasites are the cause of excessive mortalities in both the captive and wild

Published online: 21 September 2020

fishes worldwide and India is not an exception to this havoc (Yoshinaga et al. 2009; Justine et al. 2010). The ectoparasitic infection has become a major setback in the recognition and management of parasites and diseases in the quickly developing aquaculture and mariculture industry (Roza et al. 2002). Only a handful of scientists worked on the parasitic fauna of *Mystus* sp. and their community ecology (Vankara and Vijayalakshmi 2009; Vankara and Chikkam 2017; Murthy and Ratnakumari 2018). The present study will enable to evaluate the spatial distribution of the ectoparasitic fauna within the gills of *Mystus vittatus* fishes in the River Penna flowing through the YSR District.

Materials and methods

Study area

Three sampling sites of catchment areas of River Penna flowing through YSR district were selected for the host samples collected during the present study.

Site-I: Mylavaram Reservoir across the Penna River in Mylavaram village (Lat.14° 0′ 15″N 78° 20 40″ E longitude).

Site-II: Aadinimmayapalle Dam across the Penna River in Chennur Village (Lat.14″34′0.12″N, 78″48′ 0″E longitude).

Site-III: Backwaters of Somasila reservoir across the Penna River, Nellore District (14°29′22″ N 79°18′19″E) reach near Vontimitta Village.

Fish sampling

A total of 70 *Mystus vittatus* of mean (± SD) total length 7.02 ± 2.14 cm (range 3.0–11.0 cm) and mean (± SD) weight 87 ± 26.6 g (range 30.0–140.0 g) with a condition factor (25.15) were considered for the gill community structure and spatial distribution studies and 7–12 fish samples on an average were collected monthly for a period of eight months only i.e. from July 2017 to February 2018. The fish identification was done according to the keys given by Jayaram (1981) and the fishes of various sizes (small, medium, and large) were transported to the laboratory for the inspection of ectoparasites.

Morphometric analysis, collection and identification of ectoparasites

The documentation of the total length (mm), standard length (mm), and weight (g) and sex of host, along with their spectrum of ectoparasites from gills and skin was done on exclusively prepared data sheets monthly (Marcogliese 1997). The external surfaces (gills, fins, scales, eyeballs, and buccal cavity) were scrupulously observed for the presence of ectoparasites under the stereo zoom microscope (LM-52-3621 Elegant). The excised left and right gill arches were alienated from the host and kept in water-filled petridishes. Gill arches were numbered I to IV from both sides of the fish. Each arch was divided into three gill segments—dorsal, median (Anterior-median and posterior median) and ventral segments; two gill areas—proximal and distal (Turgut et al. 2006) and two gill hemibranchs—anterior (external) and posterior (internal) as a niche for parasites (Rohde 1977; Shaharom 1985). The obtained ectoparasites were observed and identified under the Lynx trinocular microscope (N-800 M). The sampled fishes were categorized into three size classes (Class-I = 4–6 cm, Class-II = 7–9 cm, Class-III = 10–12 cm) to correlate the parasitization with the size of the host. Fulton's condition factor (K-factor, $K_c = W \times 10^5/SL^3$, W is the weight (g) and SL, the standard length of fish (mm) was used to evaluate the impact of the parasites on the health condition of host fish (Klemm et al. 1993).

Statistical analysis

Jaccard's coefficient of association (r_j) is used to test the associations between all pairs of ectoparasitic species in each fish species. Coefficient equals to zero when the two ectoparasitic species in a pair never co-occur, and one when they are always found together. The interrelationship between the host length and overall parasitization was analyzed by Karl Pearson's correlation coefficient (r) and Spearman's rank correlation coefficient (r_s) (https://www.socscistatistics.com/tests/pearson/). The significance of the association between host sex and prevalence of infection was analyzed by Mann–Whitney U test (http://vassarstats.net/utest.html). Microsoft Excel 2007 and IBM SPSS 21.0 version was used to calculate the test statistics.

Data analysis

The qualitative and quantitative analysis of the data was done by using different biostatistical parameters (Sokal and Rohlf 2000) and the ecological terminology was adopted from Bush et al. (1997). Infection rate, prevalence, mean intensity, mean abundance, and index of infection were calculated for each month for ectoparasitic groups and for individual parasitic species. Dominance index, frequency distribution, and percentage of infection were calculated group-wise (i.e., for monogeneans and copepods) and species wise. All these parameters are mentioned in a tabular form and are presented graphically.

Results

Of the 70 examined host specimens, a total of 238 ectoparasites were obtained from the gills of 54 infected fishes with a prevalence of 77.1%, mean intensity of 4.4, and mean abundance of 3.4 respectively. Two species of monogeneans, *Cornudiscoides vittati* and *Bifurcohaptor indicus*, and only one species of copepod, *Lamproglena hospetensis* were obtained from the gills of *M. vittatus.* Prevalence, mean intensity, and dispersion index of the three parasitic species revealed that they have adopted an aggregation distribution (Table 1). *C. vittati* ($r_s = 0.206$, $p = 0.0866$) showed a positive but statistically insignificant correlation; *B. indicus* ($r_s = 0.361$*, $p = 0.00211$*) showed a positive and statistically significant correlation whereas *L. hospetensis* ($r_s = -0.213$, $p = 0.0764$) showed a negative and statistically insignificant correlation with the condition factor of the fish respectively (Table 2). The infestation rate of *C. vittati* and *B. indicus* was prevalent in all the three sampling sites with the highest prevalence (71.4% and 85.7% respectively) being in fishes collected from Site-I (Mylavaram) while the infestation with *L. hospetensis* was nil. The fishes of Site-II (Chennur) showed the highest infection rate (20.7%) with *L. hospetensis* (Table 3).

Spatial distribution of ectoparasites on the gills of *M. vittatus*

The prevalence and mean intensity values of the copepod, *L. hospetensis* was 12.86% and 1.66 on the right side and 8.57% and 2.3 on the left side of fish respectively (Table 4). However, the prevalence and mean intensity values of two monogeneans were 64.3% and 2.93 on the right side and 54.3% and 2.02 on the left side of fish respectively (Table 4). The prevalence and mean intensity of ectoparasitic infestation from the gill of *M. vittatus* showed that *C. vittati* was 37.1% in gill arch II, *B. indicus* was 30% in gill arch III whereas *L. hospetensis* was 12.86% in gill arch I (Table 5). Both the monogeneans and copepods shared different niches i.e., *L. hospetensis* (gill arches I), *C. vittati* (gill arch II) and *B. indicus* (gill arch III) to prevent resource competition.

Table 2 Values of Spearman's rank correlation coefficient, r_s correlating the relative condition factor and the abundance of *ectoparasites* in *Mystus vittatus*

Parasite species	r_s	*p*
Cornudiscoides vittati	0.206	0.0866
Bifurcohaptor indicus	0.361*	0.00211*
Lamproglena hospetensis	− 0.213	0.0764

*Significant level at $p < 0.05$

Correlation between body length and degree of infection

All three parasites showed different infestation rates in the different size classes with the highest prevalence for *C. vittati* in class-II; *B. indicus* showed infection in class-III and *L. hospetensis* showed infection in class-I. Student's *t* test also revealed that there is a significant difference from Size class-I and III of *C. vittati* (Student 't' = 1.99, $p = 0.0269$) and class I and III of *B. indicus* (Student't' = 2.65, $p = 0.00615$) while the classes size I and III of *L. hospetensis* (Student 't' = − 1.99, $p = 0.0269$) did not show any significant differences at $p < 0.05$ significant level (Table 6).

Correlation between host sex and degree of infection

The prevalence of monogeneans, *B. indicus* and *C. vittati* was 91.3% and 65.2% in female fishes and 53.2% and 34% in male fishes respectively. However, the prevalence of *L. hospetensis* was 21.3% male fish and only 8.7% in female fish. Host sex has no significant effect on the parasitic infection of all these parasites (Table 7). The values of Jaccard's coefficient of association (0.96) between the monogeneans, *B. indicus* and *C. vittati* showed a very strong interspecific association whereas a very meager association was observed between *L. hospetensis*, *B. indicus* and *C. vittati* (0.21, 0.26) respectively and *B. indicus* (Table 8).

Table 1 Prevalence, mean intensity, mean abundance, index of infection and dispersion index of *ectoparasites* in *Mystus vittatus*

Parasite species	Prevalence (%)	Mean intensity	Mean abundance	Index of infection	Variance	Dispersion Index (S^2/x)
Cornudiscoides vittati	17.1	2.4	0.4	0.1	4.96	2.92
Bifurcohaptor indicus	44.3	3.2	1.4	0.6	3.22	2.52
Lamproglena hospetensis	65.7	2.4	1.6	1.0	1.17	2.83

Table 3 Prevalence, mean intensity, mean abundance and index of infection of ectoparasitic infestation from the gills of *M. vittatus* collected from different locations

Collection sites	Total no. of fishes	Infected fishes	Total no. of parasites	Range	Prevalence (%)	Mean intensity	Mean abundance	Index of infection
C. vittati								
Chennur	29	13	29	1–7	44.8	2.2	1	0.4
Mylavaram	7	5	13	1–7	71.4	2.6	1.9	0.3
Somasila	34	18	48	1–7	52.9	2.7	1.4	0.7
B. indicus								
Chennur	29	18	45	1–8	62.1	2.5	1.6	1.0
Mylavaram	7	6	25	1–8	85.7	4.2	3.6	3.1
Somasila	34	17	49	1–10	50	2.9	1.4	0.7
L. hospetensis								
Chennur	29	6	13	1–4	20.7	2.2	0.4	0.1
Mylavaram	7	0	0	0	0	0	0	0
Somasila	34	6	16	1–5	17.6	2.7	0.5	0.1

Table 4 Prevalence and mean intensity of ectoparasitic infestation from the gills of *M. vittatus* in relation to host side

Parasite species	No. of fishes examined	Prevalence (%)		Mean intensity		Mean abundance		Index of infection	
		Left side	Right side	Left side	Right side	Left side	Right side	Left side	Right side
Copepods (*L. hospetensis*)	70	8.57	12.86	2.3	1.66	0.2	0.21	0.017	0.027
Monogeneans	70	54.3	64.3	2.02	2.93	1.1	1.86	0.59	1.21

Table 5 Prevalence and mean intensity of ectoparasitic infestation from the gills of *M. vittatus* as a function to the gill arch

	C. vittati				*B. indicus*				*L. hospetensis*			
	Gill arch I	Gill arch II	Gill arch III	Gill arch IV	Gill arch I	Gill arch II	Gill arch III	Gill arch IV	Gill arch I	Gill arch II	Gill arch III	Gill arch IV
No. of infected fishes	22	26	21	15	18	12	21	18	9	7	5	4
Range	1–4	1–3	1–2	1–4	1–3	1–3	1–2	1–3	1–2	0–1	1–2	0–1
No. of parasites	34	39	25	21	23	17	27	23	11	7	7	4
Prevalence	31.4	37.1	30	21.4	25.7	17.1	30	25.7	12.86	10	7.14	5.7
Mean Intensity	1.54	1.5	1.19	1.4	1.2	1.42	1.28	1.2	1.22	1	1.4	1
Mean abundance	0.48	0.55	0.35	0.3	0.32	0.24	0.38	0.32	0.16	0.1	0.1	0.06
Index of infection	0.152	0.206	0.107	0.064	0.08	0.042	0.11	0.08	0.02	0.01	0.0071	0.0033

Discussion

The present study showed both the monogeneans and copepods showed specific preferences for particular branchial arches or certain parts of the gill apparatus such as gill arches II (*C. vittati*), III (*B. indicus*), and I (*L. hospetensis*) while the least inclination towards the innermost gill arch IV which may be accredited to the strongest inflow of water currents across these sections of the gill filaments thus, making it an eventual niche for the settlement of the parasites (Turgut et al. 2006; Soylu et al. 2013; Modi and Vankara 2018). The discrepancy in water currents over the different parts of the gill surfaces can serve as a crucial factor in determining the distribution of these monogeneans and copepods (Turgut et al. 2006; Yang et al. 2006; Sujana 2015). Though there are no significant preferences

Table 6 Prevalence and mean intensity of ectoparasitic infestation from the gills of *M. vittatus* in relation to host size

Parasite species	Host length classes	No. of fishes infected	Prevalence (%)	Mean intensity	F	*p* value	Comparison two by two	t-value	*p* value
C.vittati	Class-I (n = 17)	7	41.2	2.3	2.23	0.114	Class-I–II	− 1.48	0.072
	Class-II (n = 35)	24	68.6	2.6			Class-I–III	− **1.99***	0.0269*
	Class-III (n = 18)	10	55.6	4.1			Class-II–III	− 1.09	0.139
L.hospetensis	Class-I (n = 17)	4	23.5	2.3	1.822	0.169	Class-I–II	− 0.114	0.454
	Class-II (n = 35)	8	22.9	2.5			Class-I–III	− **1.99***	0.0269*
	Class-III (n = 18)	0	0	0			Class-II–III	**1.87***	0.033*
B.indicus	Class-I (n = 17)	3	17.6	3.7	**3.60***	0.032*	Class-I–II	− 1.16	0.124
	Class-II (n = 35)	19	54.3	2.3			Class-I–III	**2.65***	0.00615*
	Class-III (n = 18)	14	77.8	2.6			Class-II–III	− **1.87***	0.033*

*Significant level at $p < 0.05$

Bold values indicate that the statisctically significant at $p < 0.05$

Table 7 Prevalence, mean intensity, mean abundance and index of infection of ectoparasitic infestation from the gills of *M. vittatus* collected in relation to host sex

Sex of species	A	B	c	Range	Prevalence (%)	Mean intensity	Mean abundance	Index of infection	Z-value
Cornudiscoides vittati									
Females	23	15	44	1–7	65.2	2.9	1.9	1.2	− 1.86, *p* = 0.063
Males	47	16	55	1–7	34	3.4	1.2	0.4	
Bifurcohaptor indicus									
Females	23	21	46	1–10	91.3	2.2	2.0	1.8	− 1.47, *p* = 0.138
Males	47	25	64	1–8	53.2	2.6	1.4	0.7	
Lamproglena hospetensis									
Females	23	2	7	2–5	8.7	3.5	0.3	0	0.775, *p* = 0.435
Males	47	10	22	1–4	21.3	2.2	0.5	0.1	

a = No. of fishes examined; b = No. of infected fishes; c = No. of parasites

*Significant level at $p < 0.05$

Table 8 Values of Jaccard's interspecific association (r_j) between each pair of ectoparasitic infestation from the gills of *M. vittatus*

Parasitic species	*B. indicus*	*C.vittati*	*L.hospetensis*
B. indicus	–	0.96	0.21
C.vittati	0.96	–	0.26
L.hospetensis	0.21	0.26	–

between right and left sides of the host for the distribution of these ectoparasites; the prevalence of these parasites was comparatively more on the right side than the left side. The preference of parasite to specific site of the host may be associated with the body symmetry of the parasites (Rohde 1993). In the present study, the preference of parasites to a specific site (dorsal segments, proximal area, and posterior hemibranchs) might be due to their protective sheltering mechanism for these ectoparasites (Wootten 1974). The aggregative type of distribution of the parasites within the hosts is an indication of the heterogeneity in their relationship and increased chances for the parasites to meet their partner for reproduction (Kennedy 1977; Poulin 1993; Combes 1995; Krasnov and Poulin 2010). In the present study, all three parasites signified the aggregated distribution pattern. There are few studies showing a significant positive correlation between parasitic abundance per host fish with the condition factor of the fish which might be due to the large size body and better health condition of the host (Poulin and Morand 2000; Yamada et al. 2008; Blahoua et al. 2016, 2017). On the other hand, some studies showed a negative and significant correlation or no correlation between monogenean parasitic abundance and condition factor of host fish (Lizama et al. 2007; Tozato 2011). The parasites, *B. indicus* also showed a positive and statistically significant correlation with the condition factor of the fish, *M. vittatus*, however, the other two parasites, *C. vittati* showed a positive but statistically insignificant correlation and *L. hospetensis* showed a negative correlation with the condition factor of the fish. The present study is correlating with the earlier studies of Blahoua et al. (2015, 2016, 2017). The parasitization within the host alters with the changes in the physico-chemical parameters of water in different locations (Modu et al. 2012). The anthropogenic activities near the water bodies will pollute the water which in turn makes the hosts more vulnerable to parasites due to weakened immune systems. In the present study, the prevalence of monogeneans was more in Site-I (Mylavaram) whereas the prevalence of copepod infestation was high in Site-II (Chennur). Many workers reported that parasitization increases with the growth of the fish (Tombi et al. 2014; Vankara and Chikkam 2015; Blahoua et al. 2016). However, Boungou et al. (2008) reported contrasting outcome that there is no influence of host size on monogenean parasitization. In the present study, though there is no significant impact of host size on parasitization the prevalence of *C. vittati* and *B. indicus* was high in the medium-sized fish and large-sized fishes respectively which might be due to the fact that the larger branchial surface area provides a great area of infestation and strongest water currents passing through gills of large-sized fishes offers a suitable condition for parasite settlement (Bilong and Tombi 2004; Tekin-Özan et al. 2008). Moreover, the larger volume of water passing through the gills of larger fish also serves as an important factor in transmitting more oncomiracidiums of monogeneans and copepodid stages of copepods into the fish (Simková et al. 2006). However, the occurrence of the copepod, *L. hospetensis* was more in small-sized fishes which may be attributed to the weak/developing immune system of the host (Koskivaara and Valtonen 1992; Alvarez-Pellitero 2008). The different feeding habits of male and female fish enable the parasites to infect them differently (Rohde 1993). In our study, host sex did not have a significant influence on the parasite fauna of *M. vittatus*, suggesting that habitat use and diet are similar for both sexes of this species.

Conclusion

These types of studies illustrate the invasion of ectoparasites and their site-specificity within the gills of a host in natural water systems. The results of this study would be an effective tool for fishery managers, biologists, and conservationists to initiate proactive management policies and regulations for the sustainable management of the outstanding stocks of this species and their parasite management in the river Penna flowing areas and surrounding ecosystems.

Acknowledgments The first author is grateful to UGC for providing financial assistance under the Faculty improvement program (FIP) with an Award No.APSC021/001(TF)ZOOLOGY/PH.DXII PLAN/ 2016-17 dt. July 2016.

Author's contribution The first author was involved in collecting the fish samples and parasites, literature collection, whereas the statistical analysis, drafting the manuscript was done by the second author who is the Research supervisor.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest related to the work.

Significance statement These studies are highly significant in establishing the uncontrolled injury caused by the ectoparasites to the gills of the host, thus distressing the fish's health ultimately. Hence, this type of prevalence and spatial distribution studies of the parasites within the host might provide inventiveness to the fish aquaculturists to execute apposite practices to eradicate the ectoparasitic infestation and make fish healthy.

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